

Faculty of Computing

Semester I, Session 2025/2026

SECJ3104 Application Development

Section 02

Project Proposal:

UTM ReMerit

Group Happy:

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Submission Date:

21/10/2025

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1.0 Executive Summary

UTM ReMerit is a mobile application designed to promote environmental awareness and responsible waste management at Universiti Teknologi Malaysia (UTM). The initiative aligns with the goals of UTM Campus Sustainability (UTMCS), which aims to develop a sustainable higher education institution (HEI) ecosystem.

Despite the university's ongoing sustainability efforts, student participation in recycling remains low due to a lack of incentives, limited awareness and inconvenience. To address these challenges, UTM ReMerit integrates technology to encourage active involvement in recycling activities. The application allows students to scan recyclable items, locate nearby recycling bins and participate in monthly campaigns to earn UTM Merit Points.

The application consists of four main subsystems:

I. **User Management Subsystem**
II. **Campaigns and Events Subsystem**
III. **SmartScanner and EcoMap Subsystem**
IV. **Analytics and Dashboard Subsystem**

The application will be developed using React, Firebase, Google Cloud Vision API, Google Maps API and Chart.js, ensuring a practical and scalable solution. It offers direct benefits to students by making recycling simple, rewarding and accessible. It also provides UTMCS with valuable data to support sustainability decision-making.

Overall, UTM ReMerit aims to build a culture of environmental responsibility within the university by connecting digital innovation with sustainable action.

2.0 Project Domain / Community Background

Universiti Teknologi Malaysia (UTM) is Malaysia's premier university in engineering and technology. As a public research university, UTM is recognised for its strong commitment to sustainability. In the Times Higher Education (THE) Impact Rankings 2023, UTM was ranked first globally for SDG 7: Affordable and Clean Energy, reflecting its dedication to advancing sustainable solutions. For example, the installation of photovoltaic systems and micro-hydro turbines to support indigenous communities.

UTM Campus Sustainability (UTMCS) coordinates the university's sustainability programs through the Living Laboratories (LL) framework. The LL approach emphasises human-centric research and development. Major LL initiatives include the Sustainable Arcade, Green Office, Sustainable Energy Management programs, etc. In addition, UTMCS also organises webinars, seminars and awareness campaigns.

Despite these efforts, a key challenge for UTM Campus Sustainability (UTMCS) is integrating sustainable practices into the daily routines of the UTM community. Although recycling facilities and awareness campaigns are in place, student participation leaves room for improvement. Factors contributing to this include a lack of consistent motivation or incentives, a limited understanding of recycling procedures and difficulty locating recycling points. These issues reduce the overall effectiveness of UTM's environmental initiatives and slow progress toward achieving a zero-waste campus.

To address these challenges, the proposed mobile application, UTM ReMerit, aims to strengthen engagement and participation among students. The application comprises four main modules: Authentication and Authorisation, Campaigns and Events, SmartScanner and EcoMap, and Analytics and Dashboard. By combining technology with environmental education, UTM ReMerit supports UTM's vision of promoting responsible and sustainable behaviour across its community.

3.0 Problem Analysis

3.1 Questionnaire

The questionnaire was created using Google Forms and distributed across various WhatsApp groups among UTM students. In the end, a total of 30 responses were obtained.

I. Demographic

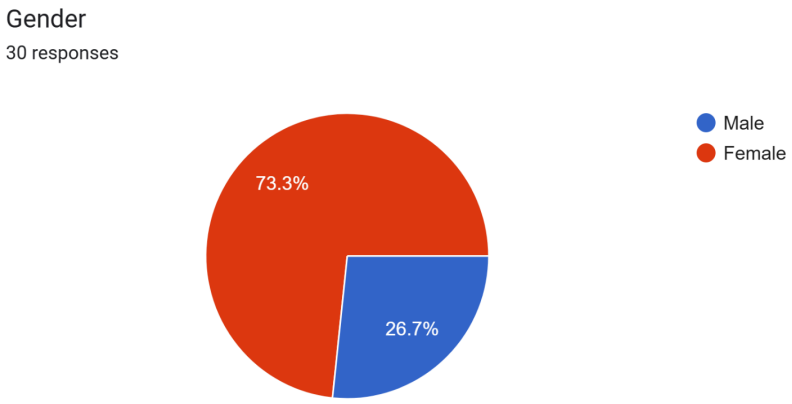


Figure 3.1.1: Pie Chart of Respondents' Gender

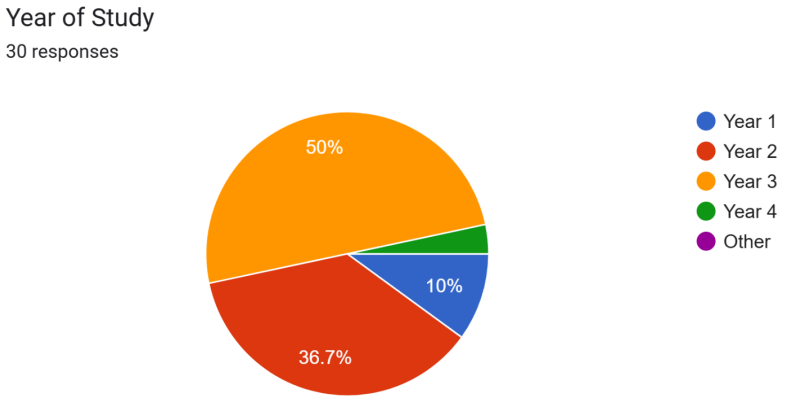


Figure 3.1.2: Pie Chart of Respondents' Year of Study

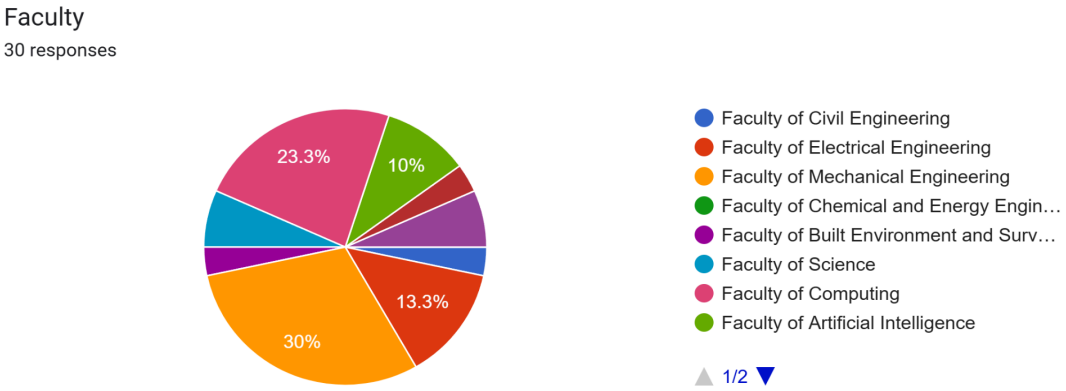


Figure 3.1.3: Pie Chart of Respondents' Faculty

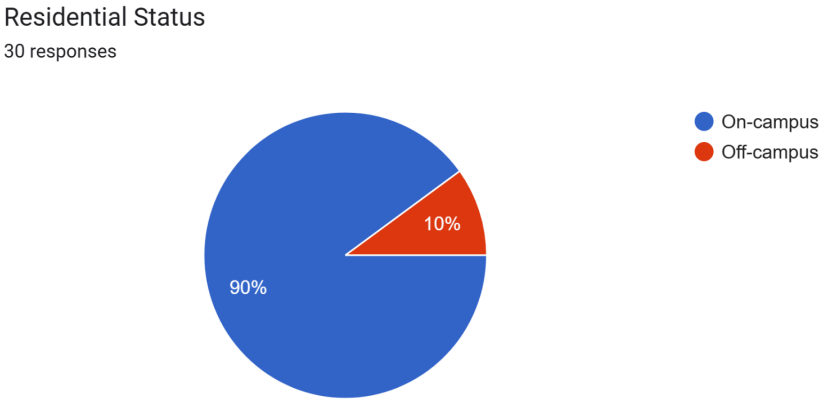


Figure 3.1.4: Pie Chart of Respondents' Residential Status

The survey gathered responses from 30 UTM students, representing various faculties and academic levels. A majority of respondents were female (73.3%) compared to male students (26.7%), and most were from Year 2 (36.7%) and Year 3 (50%), with fewer from Year 1 (10%) and Year 4 (3.3%). In terms of faculty distribution, most students came from Mechanical Engineering (30%), Computing (23.3%), and Electrical Engineering (13.3%), followed by smaller representations from Artificial Intelligence (10%), Science (6.7%), Educational Sciences and Technology (6.7%), Build Environment and Technology (3.3%), Civil Engineering (3.3%), and Management (3.3%). Additionally, 27 respondents (90%) were on-campus residents, while only 3 respondents (10%) lived off-campus. This is significant because on-campus students interact more directly with UTM's recycling facilities and sustainability initiatives.

II. Current Recycling Habits & Motivation

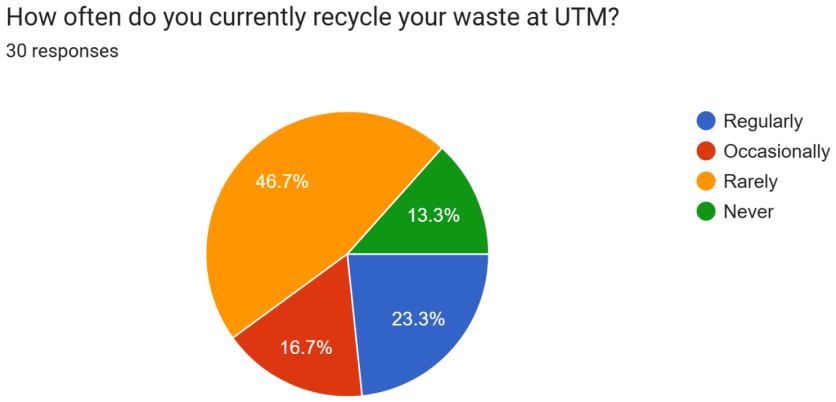


Figure 3.1.5: Pie Chart of Respondents' Recycling Frequency at UTM

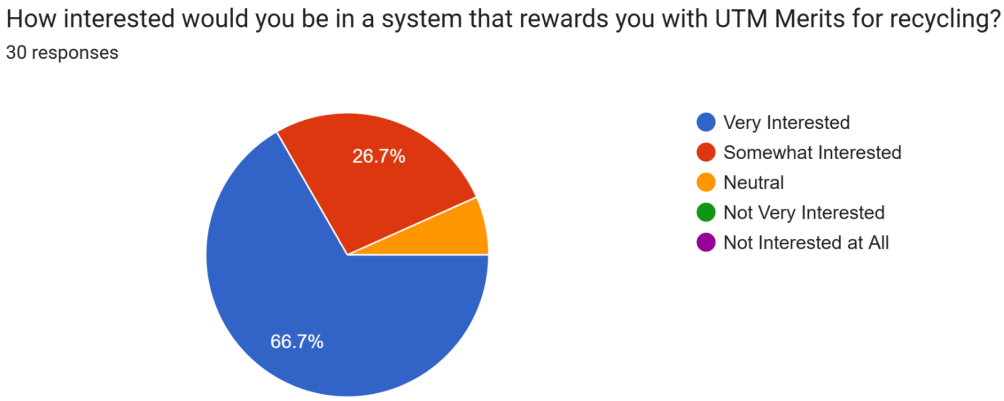


Figure 3.1.6: Pie Chart of Respondents' Interest in a System that Rewards Recycling with UTM Merit Points

Recycling frequency remains low among UTM students. When we asked about their current recycling habits, the majority of respondents (46.7%) admitted that they rarely recycle, and 13.3% never recycle their waste at UTM. Only about 23.3% recycle regularly and 16.7% occasionally recycle. Despite this, an overwhelming majority (66.7%) are very interested in a system that rewards recycling with UTM Merit Points. 26.7% were somewhat interested and 6.7% were neutral. This suggests that students are motivated but need more structured and rewarding systems to turn intention into consistent recycling behaviour.

III. Activities, Campaigns & Engagement

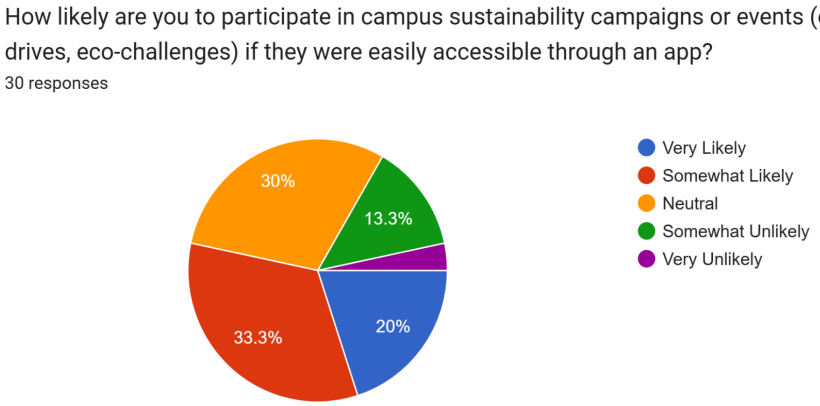


Figure 3.1.7: Pie Chart of Respondents' Likelihood to Join Sustainability Campaigns through an App

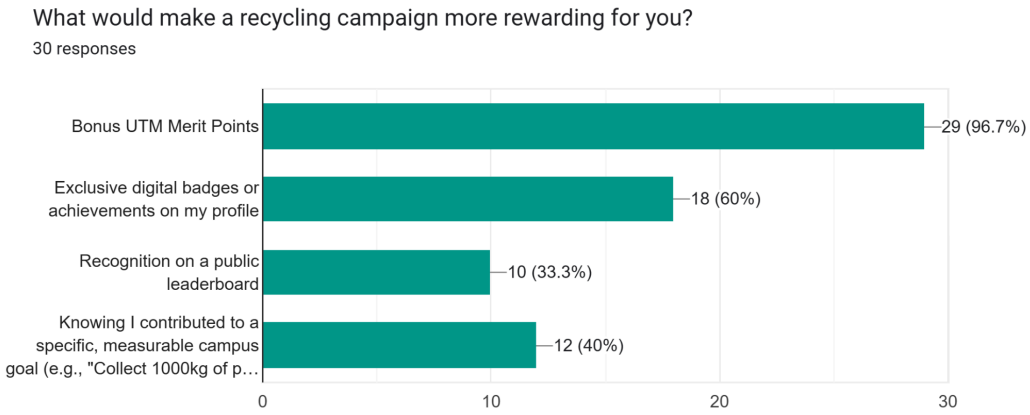


Figure 3.1.8: Bar Chart of Respondents' Preferred Motivators for Participating in Recycling Campaigns

When asked about their likelihood to join sustainability campaigns through an app, 33.3% were somewhat likely, 20.0% were very likely, while 30.0% were neutral. Only 16.6% were unlikely to participate. This indicates moderate to strong potential engagement if campaigns are easily accessible digitally. Additionally, many respondents also noted that bonus UTM Merit Points, achievement badges, and public leaderboards would make recycling campaigns more rewarding. Therefore, gamification and rewards are critical motivators for active participation in eco-related activities.

IV. Recycling Knowledge & Awareness

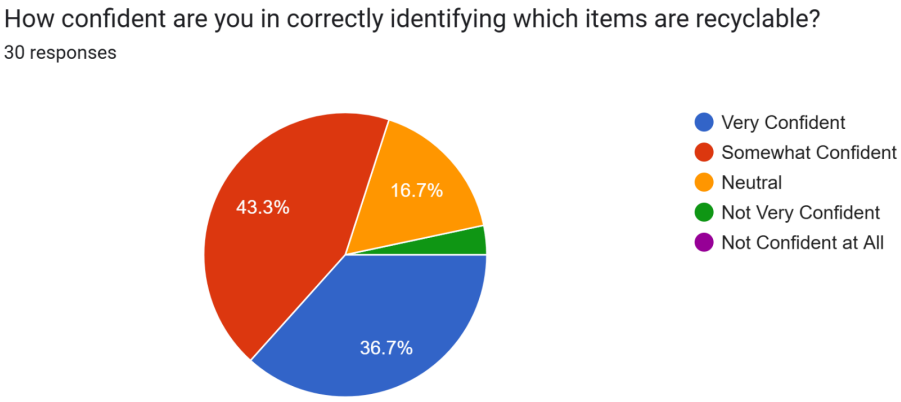


Figure 3.1.9: Pie Chart of Respondents' Confidence in Identifying Recyclable Items

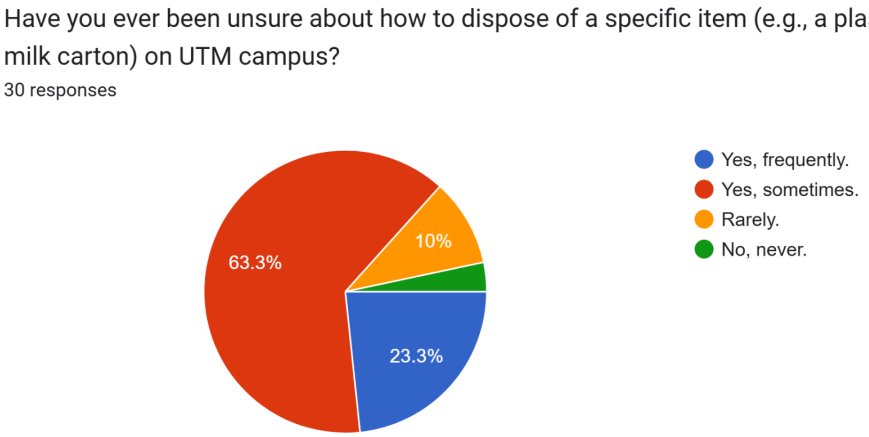


Figure 3.1.10: Pie Chart of Respondents' Frequency of Uncertainty in Disposing of Specific Items

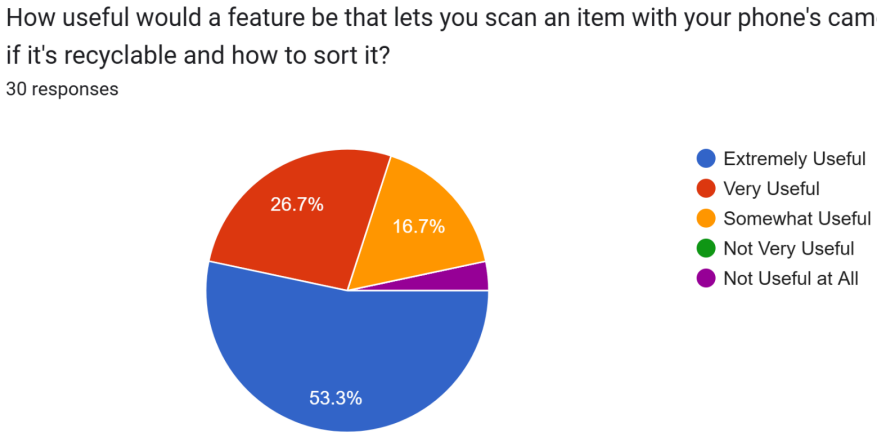


Figure 3.1.11: Pie Chart of Respondents' Perceived Usefulness of the SmartScanner Feature

In terms of confidence, 36.7% of students are very confident, and 43.3% are somewhat confident in identifying recyclable items. However, 16.7% were neutral, and 3.3% admitted to being not very confident, showing that there is still uncertainty among some students. This aligns with the finding that 63.3% of respondents have sometimes been unsure how to dispose of specific items, and 23.3% have felt this frequently. These figures clearly indicate a need for better education and tools such as the proposed SmartScanner to guide proper recycling practices.

The SmartScanner feature received overwhelming support, with 53.3% rating it as extremely useful, 26.7% as very useful, and 16.7% as somewhat useful. Only 3.3% found it not useful.
This shows that nearly all students (over 96%) perceive real value in a digital tool that identifies recyclable materials, confirming the feasibility and relevance of incorporating AI-based image recognition into the UTM ReMerit system.

V. Accessibility & Convenience

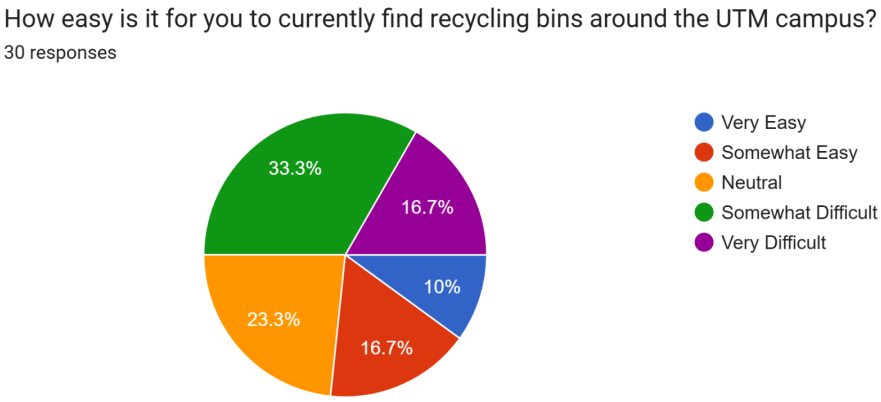


Figure 3.1.12: Pie Chart of Respondents' Difficulty Level in Locating Recycling Bins at UTM

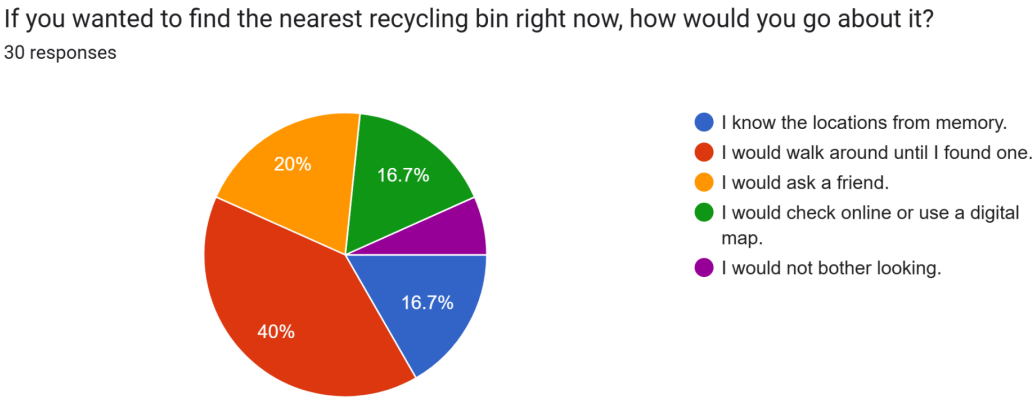


Figure 3.1.13: Bar Chart of Respondents' Usual Methods for Finding Recycling Bins

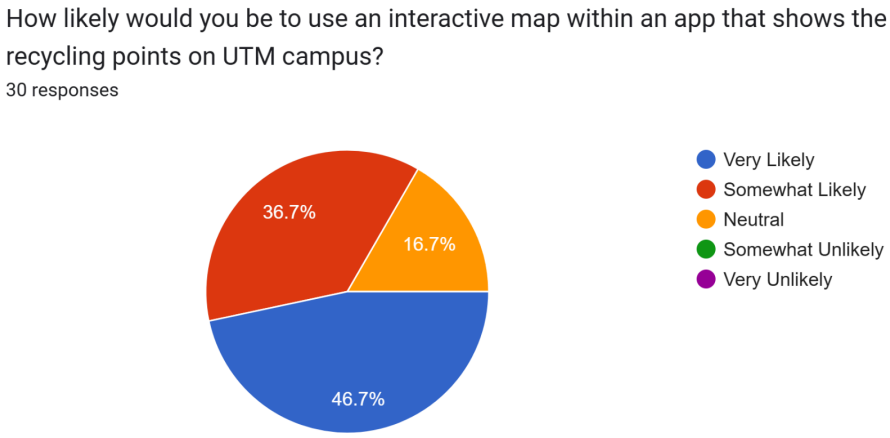


Figure 3.1.14: Pie Chart of Respondents' Likelihood to Use an Interactive Map for Recycling Bin Locations

Accessibility remains one of the major barriers to effective recycling practices among UTM students. The survey shows that 33.3% of students find it somewhat difficult and 16.7% find it very difficult to locate recycling bins, while 23.3% are neutral. Only 16.7% find it somewhat easy and 10% very easy. This shows that half of the students (50%) face difficulty in locating recycling bins on campus.

When asked how they would find a bin, most said they would walk around until they found one (40%) or ask a friend (20%), while a few mentioned they knew the locations from memory (16.7%) or would check online (16.7%). This indicates that there is currently no centralised or convenient system for locating bins.

Encouragingly, 46.7% of respondents stated they are very likely and 36.7% are somewhat likely to use an interactive map within an app that displays all recycling bin locations, while 16.7% remained neutral. These findings highlight strong support for the proposed EcoMap module, which would enhance accessibility, reduce uncertainty, and encourage more consistent recycling participation among UTM students.

4.0 Problem Statement

Despite Universiti Teknologi Malaysia's (UTM) ongoing sustainability initiatives, student participation in recycling activities remains relatively low. Findings from questionnaires indicate that several factors hinder consistent engagement and participation.

A lack of motivation and feedback remains one of the main barriers to consistent recycling behaviour among students. There is currently no mechanism for individuals to monitor their recycling performance, receive recognition or gain rewards for their efforts. The absence of progress tracking, leaderboards, or incentives such as UTM Merit Points reduces sustained engagement and interest.

Uncertainty about proper recycling practices also contributes to low participation. Many students are unsure which materials are recyclable or how to sort waste correctly. This lack of understanding results in improper disposal and decreases the overall effectiveness of campus recycling programs.

Accessibility further affects participation levels. Students often report difficulty locating recycling bins or are unaware of their exact locations. The lack of clear directions or a centralised system to guide users creates inconvenience and discourages consistent recycling habits.

In addition, there is room for improvement in tracking and analysing recycling data across the campus. Improved data collection and analytical tools would allow for better monitoring and more informed decision-making.

Overall, these challenges highlight the need for an integrated, technology-driven approach that addresses motivation, awareness, accessibility and data management to promote more effective recycling behaviour among UTM students and advance the university's goal of a zero-waste campus.

5.0 Goal / Objectives

The goal of this project is to develop UTM ReMerit, a mobile application that promotes sustainable recycling practices among UTM students by improving motivation, awareness and accessibility.

The objectives of this project are as follows:

I. To enhance motivation and participation through progress tracking and rewards

Students often lack motivation because their recycling efforts are not visible or directly rewarded. This objective focuses on allowing users to track their recycling progress and view leaderboards. Upon reaching a minimum contribution during monthly campaigns, users may qualify for UTM Merit Points, providing a meaningful incentive for continued participation.

II. To increase recycling awareness and knowledge

Many students are uncertain about proper recycling practices or which materials are recyclable. Through the AI-powered camera feature, users can scan and identify recyclable items.

III. To improve accessibility and convenience in recycling

Many students are unaware of recycling bin locations or find them difficult to access. This objective aims to provide an interactive map that displays nearby recycling points across the campus.

IV. To support UTM Campus Sustainability (UTMCS) in monitoring and evaluation

This objective aims to assist UTMCS in collecting and analysing recycling data efficiently. Insights on user participation and recycling trends will enable the university to plan and improve initiatives based on data and information.

6.0 Proposed Application

The proposed mobile application, UTM ReMerit, allows users to identify recyclable items, locate recycling bins, track their recycling performance and earn rewards that can be converted into UTM Merit Points. The application consists of four main subsystems, with 3 modules each.

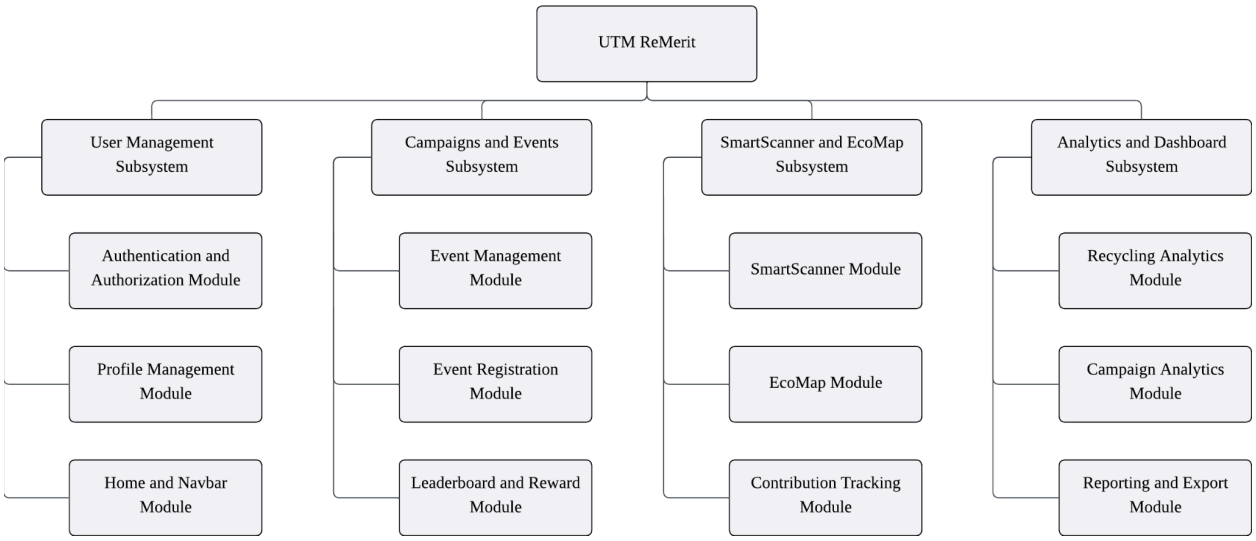


Figure 6.0.1: Overview of UTM ReMerit Subsystems and Modules

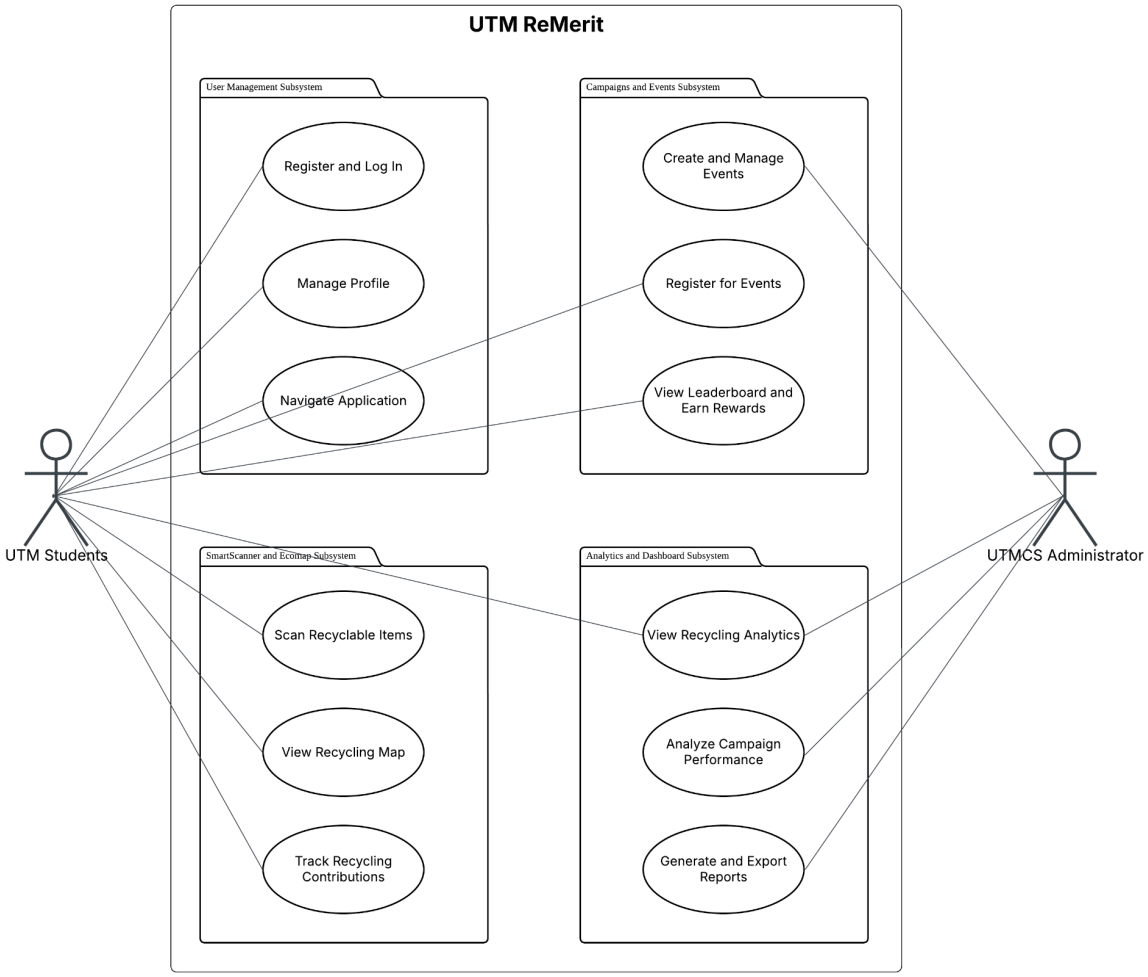


Figure 6.0.2: Use Case Diagram of UTM ReMerit System

6.1 User Management Subsystem

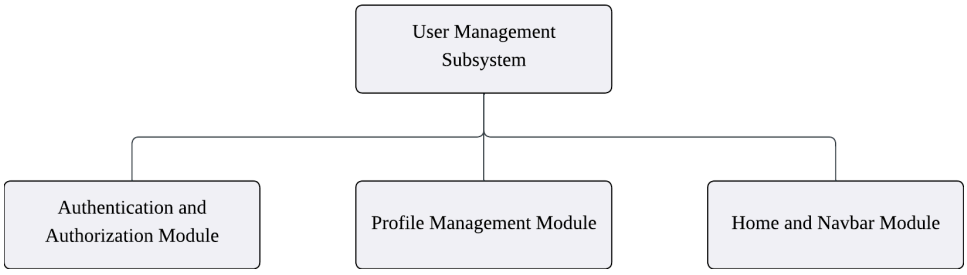


Figure 6.1.1: Overview of User Management Subsystem

This subsystem is responsible for ensuring secure access and personalised interaction within the application. It consists of the Authentication and Authorisation Module, Profile Management Module, and Homepage and NavBar Module.

6.1.1 Authentication and Authorisation Module

This module manages user registration, login and role-based access control. It ensures that only authorised users can access the system and protects user data to maintain security and system integrity.

6.1.2 Profile Management Module

This module stores and manages user information and personal details. It allows users to view and update their profiles.

6.1.3 Homepage and Navigation Bar Module

This module provides users with clear and intuitive access to the main features of the application. It serves as the central interface for navigation.

6.2 Campaigns and Events Subsystem

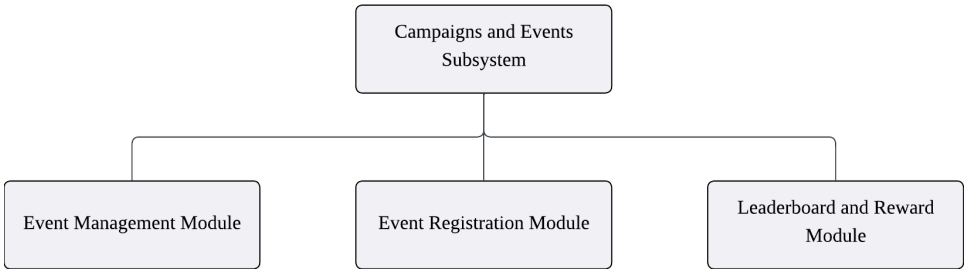


Figure 6.2.1: Overview of Campaigns and Events Subsystem

This subsystem manages user engagement through sustainability campaigns and recycling-related events. It consists of the Event Management Module, Event Registration Module, and Leaderboard and Reward Module.

6.2.1 Event Management Module

This module allows administrators to create, update and manage campaigns and events.

6.2.2 Event Registration Module

This module enables students to browse available events, register for participation and track their involvement. It simplifies the registration process and encourages active engagement in sustainability activities.

6.2.3 Leaderboard and Reward Module

This module displays user rankings based on recycling performance. It motivates participation by showcasing top contributors and rewarding users with UTM Merit Points.

6.3 SmartScanner and EcoMap Subsystem

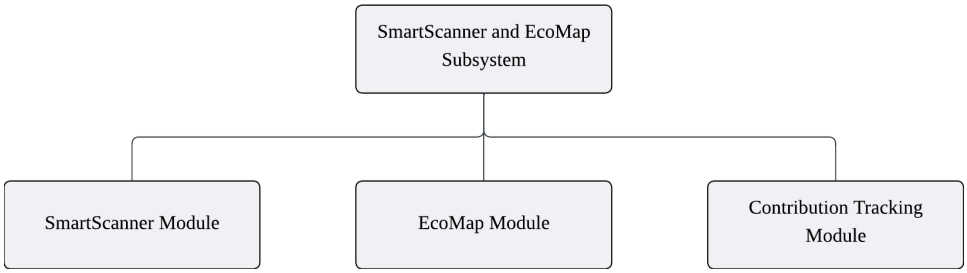


Figure 6.3.1: Overview of SmartScanner and EcoMap Subsystem

This subsystem streamlines recycling operations by helping users identify recyclables and locate recycling facilities. It consists of the SmartScanner Module, EcoMap Module and Contribution Tracking Module.

6.3.1 SmartScanner Module

This module uses AI-based image recognition or QR scanning to identify recyclable materials. It helps users determine the correct disposal method and promotes proper recycling habits.

6.3.2 EcoMap Module

This module provides an interactive map displaying recycling bins and drop-off points around the campus. It guides users to the nearest available location and improves accessibility and convenience.

6.3.3 Contribution Tracking Module

This module logs recycling activities and monitors user contributions. It provides accurate tracking of performance and supports the merit-based reward system.

6.4 Analytics and Dashboard Subsystem

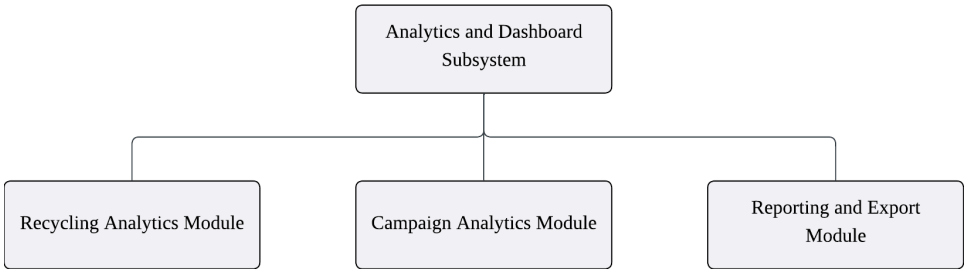


Figure 6.4.1: Overview of Analytics and Dashboard Subsystem

This subsystem supports data monitoring, analysis and reporting for both students and administrators. It consists of the Recycling Analytics Module, Campaign Analytics Module, and Reporting and Export Module.

6.4.1 Recycling Analytics Module

This module visualises individual and collective recycling performance through charts and statistics. It helps users understand their progress and contribution to sustainability goals.

6.4.2 Campaign Analytics Module

This module analyses participation data and evaluates the effectiveness of sustainability campaigns. It provides administrators with insights for improving future initiatives.

6.4.3 Reporting and Export Module

This module generates summary reports on recycling data and campaign performance, helping UTM Campus Sustainability (UTMCS) monitor progress and make informed decisions.

7.0 Proposed Technology and Tools Used

Table 7.0.1: Proposed Technology and Tools Used

Component	Technology / Tool	Purpose and Usage
Frontend	React Native	Enables cross-platform (Android and iOS) mobile application development using JavaScript.
	Expo	Simplifies development and testing by allowing real-time preview of the app on mobile devices through QR code scanning. In addition, Expo provides built-in access to camera and location services.
Backend	Firebase Authentication	Handles secure user login, registration and role-based access without manual server setup.
	Firebase Firestore	A NoSQL cloud database for storing user profiles, recycling records, campaign details and analytics data in real time.
	Firebase Cloud Functions	Manages backend logic such as leaderboard generation and merit point updates.
Artificial Intelligence (AI)	Google Cloud Vision API	Performs accurate image recognition to identify recyclable materials when users scan items. This cloud-based API removes the need for model training and ensures consistent performance.
Mapping and Location Services	Google Maps API	Provides geolocation features that display nearby recycling bins and drop-off points within the EcoMap module.
Analytics and Visualisation	Chart.js	Used to visualise analytics in the form of clear, interactive visual reports.
Development Environment	Android Studio / Visual Studio Code	Used for coding, debugging and testing the mobile application across various screen sizes and devices.
Project Management	Trello	Supports sprint-based project management and progress tracking to ensure timely development.

Version Control	GitHub	Manages version control and facilitates collaboration between team members by maintaining a shared repository for the source code.

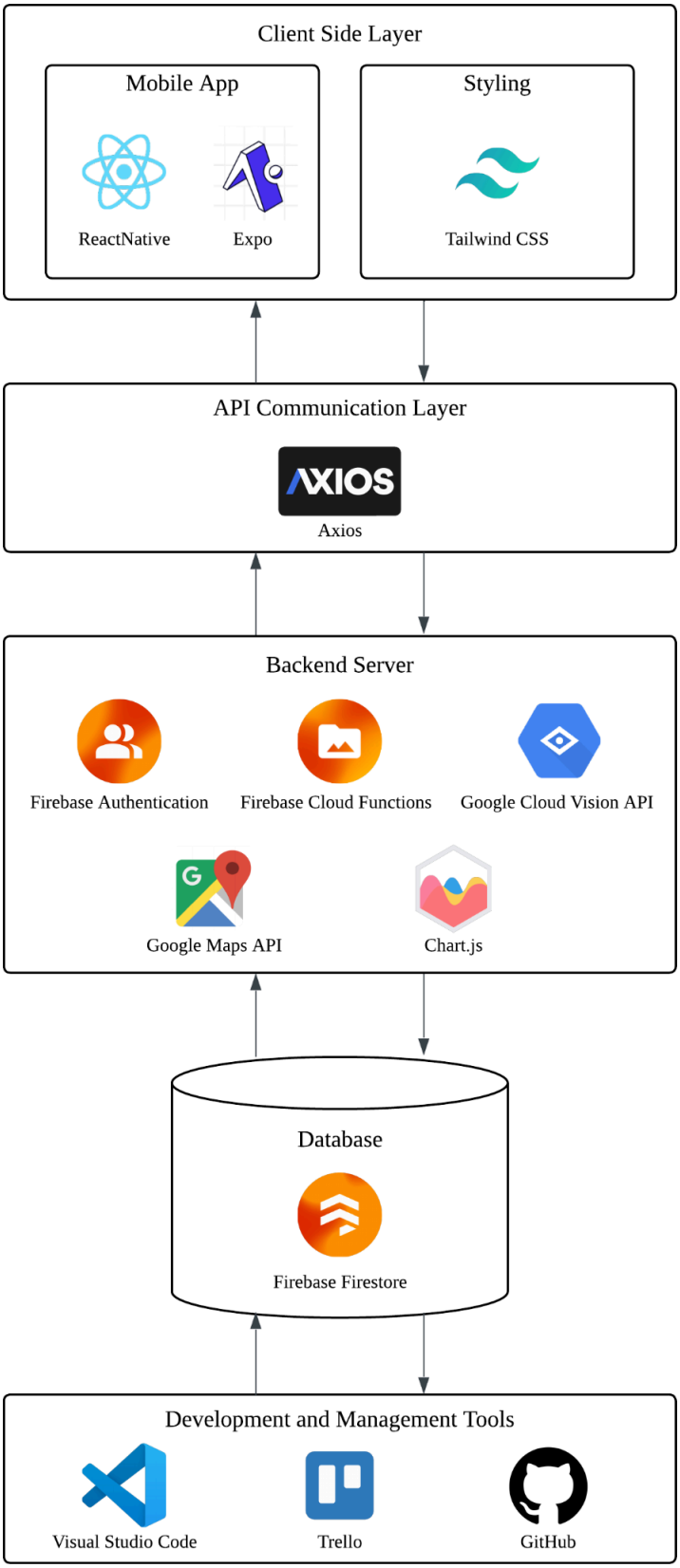


Figure 7.0.1: System Architecture Diagram of UTM ReMerit

8.0 Project Benefit

The UTM ReMerit application provides several key benefits that align with the project goals and system modules.

Firstly, the application increases motivation and participation in recycling activities. Through the Leaderboard and Reward Module in the Campaigns and Events Subsystem, users can track their progress and earn UTM Merit Points. This supports the goal of encouraging consistent participation through progress tracking and rewards.

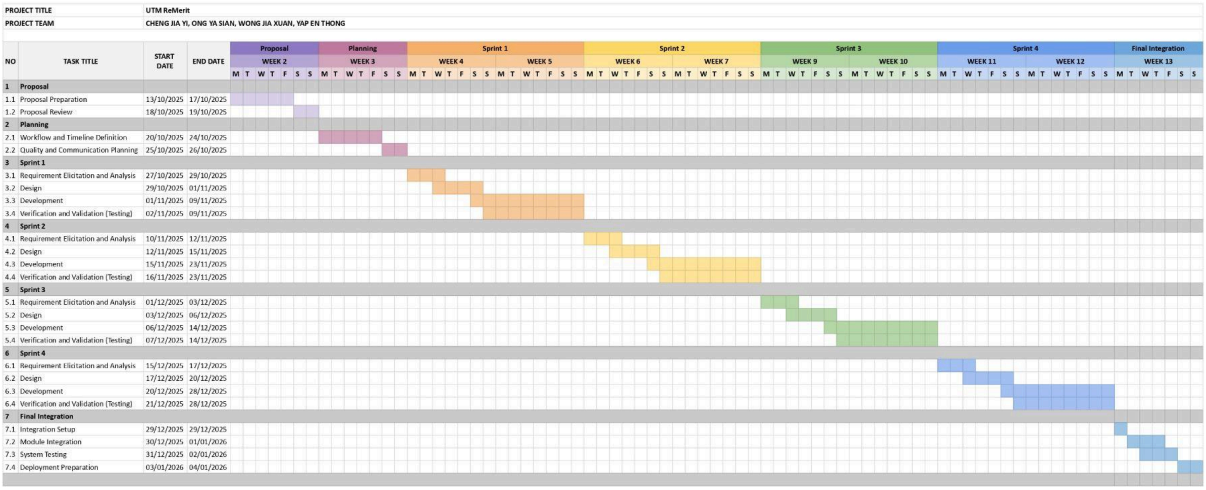
Secondly, the SmartScanner Module in the SmartScanner and EcoMap Subsystem helps users identify recyclable materials and learn proper recycling practices. This directly supports the goal of increasing recycling awareness and knowledge among students.

Thirdly, the EcoMap Module improves accessibility by showing the locations of recycling bins and drop-off points around the campus. This supports the goal of making recycling easier and more convenient for users.

The Analytics and Dashboard Subsystem benefits UTM Campus Sustainability (UTMCS) by collecting and analysing recycling data. The Recycling Analytics, Campaign Analytics, and Reporting and Export Modules allow administrators to monitor participation and campaign effectiveness. This supports the goal of improving sustainability monitoring and decision-making.

Overall, UTM ReMerit contributes to the creation of a more sustainable and responsible campus community. The application not only encourages students to take part in recycling activities but also provides administrators with accurate data to improve sustainability initiatives. UTM ReMerit supports UTM's long-term objective of promoting environmental responsibility and achieving a low-carbon, zero-waste campus.

9.0 Project Schedule



PROJECT TITLE	UTM ReMerit
PROJECT TEAM	CHENG JIA YI, ONG YA SIAN, WONG JIA XUAN, YAP EN THONG

NO	TASK TITLE	START DATE	END DATE
1	**Proposal**		
1.1	Proposal Preparation	13/10/2025	17/10/2025
1.2	Proposal Review	18/10/2025	19/10/2025
2	**Planning**		
2.1	Workflow and Timeline Definition	20/10/2025	24/10/2025
2.2	Quality and Communication Planning	25/10/2025	26/10/2025
3	**Sprint 1**		
3.1	Requirement Elicitation and Analysis	27/10/2025	29/10/2025
3.2	Design	29/10/2025	01/11/2025
3.3	Development	01/11/2025	09/11/2025
3.4	Verification and Validation (Testing)	02/11/2025	09/11/2025
4	**Sprint 2**		
4.1	Requirement Elicitation and Analysis	10/11/2025	12/11/2025
4.2	Design	12/11/2025	15/11/2025
4.3	Development	15/11/2025	23/11/2025
4.4	Verification and Validation (Testing)	16/11/2025	23/11/2025
5	**Sprint 3**		
5.1	Requirement Elicitation and Analysis	01/12/2025	03/12/2025
5.2	Design	03/12/2025	06/12/2025
5.3	Development	06/12/2025	14/12/2025
5.4	Verification and Validation (Testing)	07/12/2025	14/12/2025
6	**Sprint 4**		
6.1	Requirement Elicitation and Analysis	15/12/2025	17/12/2025
6.2	Design	17/12/2025	20/12/2025
6.3	Development	20/12/2025	28/12/2025
6.4	Verification and Validation (Testing)	21/12/2025	28/12/2025
7	**Final Integration**		
7.1	Integration Setup	29/12/2025	29/12/2025
7.2	Module Integration	30/12/2025	01/01/2026
7.3	System Testing	31/12/2025	02/01/2026
7.4	Deployment Preparation	03/01/2026	04/01/2026

Figure 9.0.1: Gantt Chart for the Overall Project Development

10.0 Team Members and Roles

10.1 Distribution of Scrum Roles

The Scrum roles utilised in this project are as follows:

I. Project Manager

The Project Manager is responsible for ensuring that the project objectives are achieved according to the specified requirements. This role involves managing the project scope, monitoring progress and ensuring that all deliverables meet quality standards and deadlines.

II. Scrum Master

The Scrum Master facilitates the Scrum process and ensures that the team follows agile principles. This role involves organising sprint meetings, removing obstacles that may hinder progress and maintaining effective communication among team members.

III. Developer

The Developer is responsible for implementing the system features and functions according to the project requirements. This includes coding, integrating system components and ensuring that the application performs as expected.

IV. Tester

The Tester is responsible for verifying and validating the functionality of the system. This includes identifying and reporting bugs, conducting tests to ensure system reliability

Table 10.1.1: Distribution of Scrum Roles during each Sprint

Role	**Name**
Sprint 1	
Product Manager	Cheng Jia Yi
Scrum Master	Ong Ya Sian
Developer	Wong Jia Xuan
Tester	Yap En Thong
Sprint 2	
Product Manager	Ong Ya Sian
Scrum Master	Wong Jia Xuan
Developer	Yap En Thong
Tester	Cheng Jia Yi
Sprint 3	
Product Manager	Wong Jia Xuan
Scrum Master	Yap En Thong
Developer	Cheng Jia Yi
Tester	Ong Ya Sian
Sprint 4	
Product Manager	Yap En Thong
Scrum Master	Cheng Jia Yi
Developer	Ong Ya Sian
Tester	Wong Jia Xuan

10.2 Distribution of Subsystems and Modules

Table 10.2.1: Distribution of Subsystems and Modules during each Sprint

Subsystem	Module	Name
Sprint 1		
User Management	Authentication and Authorisation	Cheng Jia Yi
Campaigns and Events	Event Management	Ong Ya Sian
SmartScanner and EcoMap	SmartScanner	Wong Jia Xuan
Analytics and Dashboard	Recycling Analytics	Yap En Thong
Sprint 2		
User Management	Profile Management	Ong Ya Sian
Campaigns and Events	Event Registration	Wong Jia Xuan
SmartScanner and EcoMap	EcoMap	Yap En Thong
Analytics and Dashboard	Campaign Analytics	Cheng Jia Yi
Sprint 3		
User Management	Home and Navbar	Wong Jia Xuan
Campaigns and Events	Leaderboard and Reward	Yap En Thong
SmartScanner and EcoMap	Contribution Tracking	Cheng Jia Yi
Analytics and Dashboard	Reporting and Export	Ong Ya Sian

11.0 Individual Schedule

The following Gantt charts illustrate the timeline and responsibilities for each member across all sprints.

PROJECT TITLE	UTM ReMerit
PROJECT MANAGER	CHENG JIA YI

NUMBER	TASK TITLE	START DATE	END DATE
1	**User Management Subsystem**		
	Module 1: Authentication and Authorization Module		
1.1	Requirement Elicitation and Analysis	27/10/2025	28/10/2025
1.2	Design	29/10/2025	31/10/2025
1.3	Development		
1.3.1	Frontend	01/11/2025	08/11/2025
1.3.2	Backend	02/11/2025	08/11/2025
1.3.3	Exception Handling	03/11/2025	08/11/2025
1.4	Verification and Validation (Testing)	02/11/2025	09/11/2025
2	**Analytics and Dashboard Subsystem**		
	Module 2: Campaign Analytics Module		
2.1	Requirement Elicitation and Analysis	10/11/2025	11/11/2025
2.2	Design	12/11/2025	14/11/2025
2.3	Development		
2.3.1	Frontend	15/11/2025	22/11/2025
2.3.2	Backend	16/11/2025	22/11/2025
2.3.3	Exception Handling	17/11/2025	22/11/2025
2.4	Verification and Validation (Testing)	16/11/2025	23/11/2025
3	**SmartScanner and EcoMap Subsystem**		
	Module 3: Contribution Tracking Module		
3.1	Requirement Elicitation and Analysis	01/12/2025	02/12/2025
3.2	Design	03/12/2025	05/12/2025
3.3	Development		
3.3.1	Frontend	06/12/2025	13/12/2025
3.3.2	Backend	07/12/2025	13/12/2025
3.3.3	Exception Handling	8/12/2025	13/12/2025
3.4	Verification and Validation (Testing)	07/12/2025	14/12/2025
4	**Integration**		
4.1	Requirement Elicitation and Analysis	15/12/2025	16/12/2025
4.2	Design	17/12/2025	19/12/2025
4.3	Development		
4.3.1	Frontend	20/12/2025	27/12/2025
4.3.2	Backend	21/12/2025	27/12/2025
4.3.3	Exception Handling	22/12/2025	27/12/2025
4.4	Verification and Validation (Testing)	21/12/2025	28/12/2025

Figure 11.0.1: Cheng Jia Yi's Gantt Chart

PROJECT TITLE	UTM ReMerit
PROJECT MANAGER	ONG YA SIAN

NUMBER	TASK TITLE	START DATE	END DATE
1	**Campaigns and Events Subsystem**		
	Module 1: Event Management Module		
1.1	Requirement Elicitation and Analysis	27/10/2025	28/10/2025
1.2	Design	29/10/2025	31/10/2025
1.3	Development		
1.3.1	Frontend	01/11/2025	08/11/2025
1.3.2	Backend	01/11/2025	08/11/2025
1.3.3	Exception Handling	01/11/2025	08/11/2025
1.4	Verification and Validation (Testing)	05/11/2025	09/11/2025
2	**User Management Subsystem**		
	Module 2: Profile Management Module		
2.1	Requirement Elicitation and Analysis	10/11/2025	11/11/2025
2.2	Design	12/11/2025	14/11/2025
2.3	Development		
2.3.1	Frontend	15/11/2025	22/11/2025
2.3.2	Backend	15/11/2025	22/11/2025
2.3.3	Exception Handling	15/11/2025	22/11/2025
2.4	Verification and Validation (Testing)	19/11/2025	23/11/2025
3	**Analytics and Dashboard Subsystem**		
	Module 3: Reporting and Export Module		
3.1	Requirement Elicitation and Analysis	01/12/2025	02/12/2025
3.2	Design	03/12/2025	05/12/2025
3.3	Development		
3.3.1	Frontend	06/12/2025	13/12/2025
3.3.2	Backend	06/12/2025	13/12/2025
3.3.3	Exception Handling	06/12/2025	13/12/2025
3.4	Verification and Validation (Testing)	10/12/2025	14/12/2025
4	**Integration**		
4.1	Requirement Elicitation and Analysis	15/12/2025	16/12/2025
4.2	Design	17/12/2025	19/12/2025
4.3	Development		
4.3.1	Frontend	20/12/2025	27/12/2025
4.3.2	Backend	20/12/2025	27/12/2025
4.3.3	Exception Handling	20/12/2025	27/12/2025
4.4	Verification and Validation (Testing)	24/12/2025	28/12/2025

Figure 11.0.2: Ong Ya Sian's Gantt Chart

PROJECT TITLE	UTM ReMerit
PROJECT MANAGER	WONG JIA XUAN

NUMBER	TASK TITLE	START DATE	END DATE
1	**SmartScanner and EcoMap Subsystem**		
	Module 1: SmartsScanner Module		
1.1	Requirement Elicitation and Analysis	27/10/2025	29/10/2025
1.2	Design	30/10/2025	01/11/2025
1.3	Development		
1.3.1	Frontend	02/11/2025	08/11/2025
1.3.2	Backend	02/11/2025	08/11/2025
1.3.3	Exception Handling	06/11/2025	09/11/2025
1.4	Verification and Validation (Testing)	06/11/2025	09/11/2025
2	**Campaigns and Events Subsystem**		
	Module 2: Event Registration Module		
2.1	Requirement Elicitation and Analysis	10/11/2025	12/11/2025
2.2	Design	13/11/2025	15/11/2025
2.3	Development		
2.3.1	Frontend	16/11/2025	22/11/2025
2.3.2	Backend	16/11/2025	22/11/2025
2.3.3	Exception Handling	20/11/2025	23/11/2025
2.4	Verification and Validation (Testing)	20/11/2025	23/11/2025
3	**User Management Subsystem**		
	Module 3: Home and NavBar Module		
3.1	Requirement Elicitation and Analysis	01/12/2025	03/12/2025
3.2	Design	04/12/2025	06/12/2025
3.3	Development		
3.3.1	Frontend	7/12/2025	13/12/2025
3.3.2	Backend	7/12/2025	13/12/2025
3.3.3	Exception Handling	11/12/2025	14/12/2025
3.4	Verification and Validation (Testing)	11/12/2025	14/12/2025
4	**Integration**		
4.1	Requirement Elicitation and Analysis	15/12/2025	17/12/2025
4.2	Design	18/12/2025	20/12/2025
4.3	Development		
4.3.1	Frontend	21/12/2025	27/12/2025
4.3.2	Backend	21/12/2025	27/12/2025
4.3.3	Exception Handling	25/12/2025	28/12/2025
4.4	Verification and Validation (Testing)	25/12/2025	28/12/2025

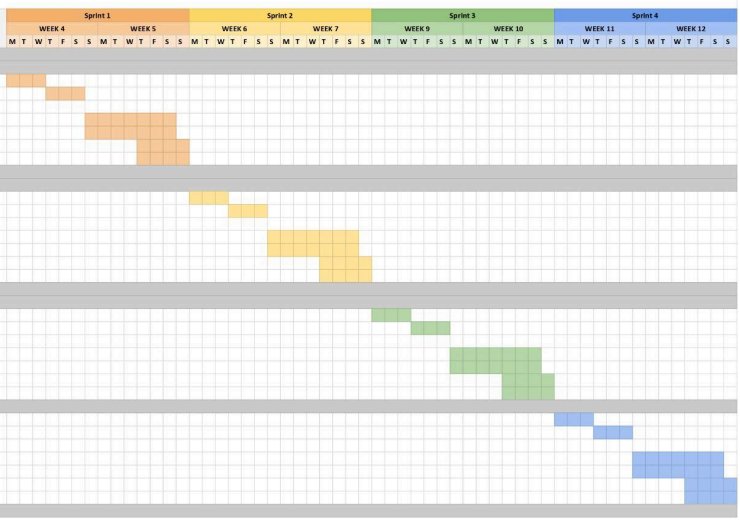


Figure 11.0.3: Wong Jia Xuan's Gantt Chart

PROJECT TITLE	UTM ReMerit
PROJECT MANAGER	YAP EN THONG

NUMBER	TASK TITLE	START DATE	END DATE
1	**Analytics and Dashboard Subsystem**		
	Module 1: Recycling Analytics Module		
1.1	Requirement Elicitation and Analysis	27/10/2025	28/10/2025
1.2	Design	29/10/2025	31/10/2025
1.3	Development		
1.3.1	Frontend	01/11/2025	08/11/2025
1.3.2	Backend	02/11/2025	08/11/2025
1.3.3	Exception Handling	03/11/2025	08/11/2025
1.4	Verification and Validation (Testing)	03/11/2025	09/11/2025
2	**SmartScanner and EcoMap Subsystem**		
	Module 2: EcoMap Module		
2.1	Requirement Elicitation and Analysis	10/11/2025	11/11/2025
2.2	Design	12/11/2025	14/11/2025
2.3	Development		
2.3.1	Frontend	15/11/2025	22/11/2025
2.3.2	Backend	16/11/2025	22/11/2025
2.3.3	Exception Handling	17/11/2025	22/11/2025
2.4	Verification and Validation (Testing)	17/11/2025	23/11/2025
3	**Campaigns and Events Subsystem**		
	Module 3: Leaderboard and Module		
3.1	Requirement Elicitation and Analysis	01/12/2025	02/12/2025
3.2	Design	03/12/2025	05/12/2025
3.3	Development		
3.3.1	Frontend	06/12/2025	13/12/2025
3.3.2	Backend	07/12/2025	13/12/2025
3.3.3	Exception Handling	08/12/2025	13/12/2025
3.4	Verification and Validation (Testing)	08/12/2025	14/12/2025
4	**Integration**		
4.1	Requirement Elicitation and Analysis	15/12/2025	16/12/2025
4.2	Design	17/12/2025	19/12/2025
4.3	Development		
4.3.1	Frontend	20/12/2025	27/12/2025
4.3.2	Backend	21/12/2025	27/12/2025
4.3.3	Exception Handling	22/12/2025	27/12/2025
4.4	Verification and Validation (Testing)	22/12/2025	28/12/2025

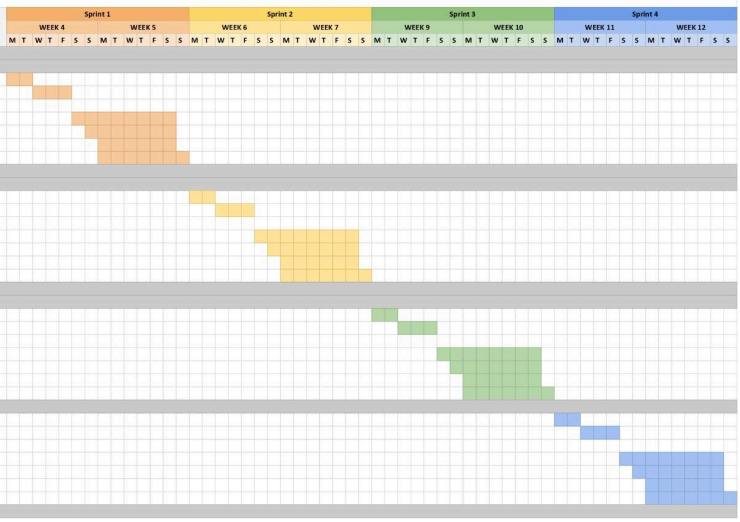


Figure 11.0.4: Yap En Thong's Gantt Chart